

Derivative Rules

1. Constant Rule: $\frac{d}{dx}[c] = \underline{\hspace{2cm}}$
2. Constant Multiple Rule: $\frac{d}{dx}[cf(x)] = \underline{\hspace{2cm}}$
3. Power Rule: $\frac{d}{dx}[x^n] = \underline{\hspace{2cm}}$
4. Sum and Difference Rules: $\frac{d}{dx}[f(x) \pm g(x)] = \underline{\hspace{2cm}}$
5. Product Rule: $\frac{d}{dx}[f(x) \cdot g(x)] = \underline{\hspace{2cm}}$
6. Quotient Rule: $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \underline{\hspace{2cm}}$
7. Chain Rule: $\frac{d}{dx}[f(g(x))]$ = $\underline{\hspace{2cm}}$
8. $\frac{d}{dx}[e^x] = \underline{\hspace{2cm}}$
9. $\frac{d}{dx}[a^x] = \underline{\hspace{2cm}}$
10. $\frac{d}{dx}[\ln x] = \underline{\hspace{2cm}}$
11. $\frac{d}{dx}[\log_a x] = \underline{\hspace{2cm}}$
12. $\frac{d}{dx}[\sin x] = \underline{\hspace{2cm}}$
13. $\frac{d}{dx}[\cos x] = \underline{\hspace{2cm}}$
14. $\frac{d}{dx}[\tan x] = \underline{\hspace{2cm}}$
15. $\frac{d}{dx}[\csc x] = \underline{\hspace{2cm}}$
16. $\frac{d}{dx}[\sec x] = \underline{\hspace{2cm}}$
17. $\frac{d}{dx}[\cot x] = \underline{\hspace{2cm}}$
18. $\frac{d}{dx}[\sin^{-1} x] = \underline{\hspace{2cm}}$
19. $\frac{d}{dx}[\cos^{-1} x] = \underline{\hspace{2cm}}$
20. $\frac{d}{dx}[\tan^{-1} x] = \underline{\hspace{2cm}}$